Causality Assessment in Drug Induced Liver Injury

FDA, PhRMA, AASLD Symposium

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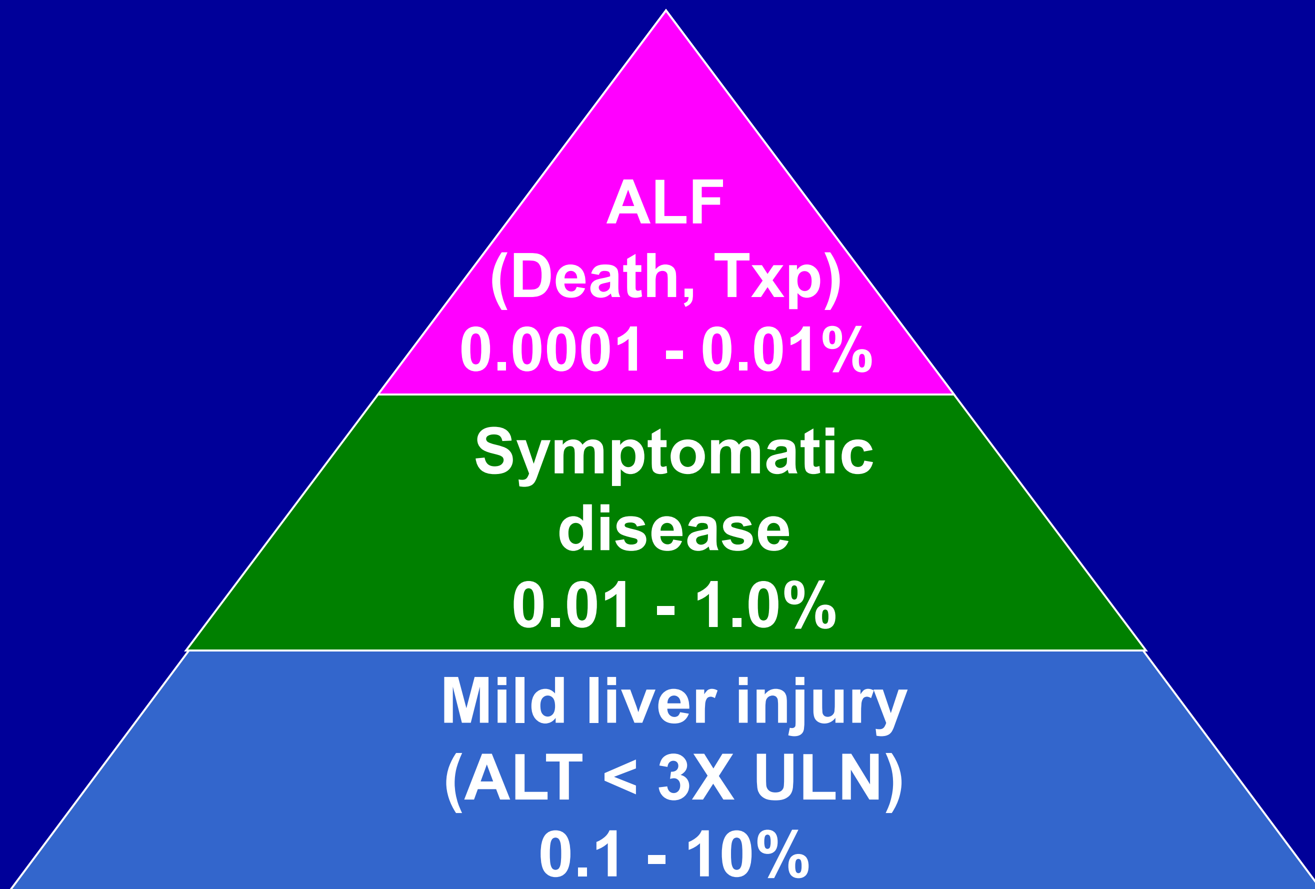
DILI- Causality Assessment

- Epidemiology
  - Differential diagnosis
- Causality instruments
  - RUCAM
  - CDS
- DILIN network

Spectrum of DILI



DILI Population based study
81,300 French ’97-’00

- 95 suspected cases
  - 34 probable DILI
    - 25% antibiotics 23% psychotropic 13% antilipid
  - 80% outpatients
  - 2 (7%) deaths

- Incidence: 14 to 24 per 100,000
  - 8,000 annual cases, 500 deaths
  - 16 X > than ADR surveillance

(Hepatology 2002; 36)

DILI Manifestations

- Acute hepatitis
- Acute cholestasis

- Chronic hepatitis
- Fatty liver/ NASH
- Granulomatous disease
- Fibrosis/ cirrhosis
- Vanishing bile duct
- VOD, peliosis
- Benign & malignant neoplasia

DILI Diagnosis

- Temporal relationship
  - Not dose related
  - ? Clinical risk factors
- Biochemical injury pattern
  - "Signature" vs protean
  - Prior reports/ cases
  - Exclude other likely causes
- Improvement with discontinuation

Acute DILI

- **Hepatocellular:** R > 5 and ALT > 2x ULN or baseline
- **Cholestatic:** R < 2 and Alk > ULN
- **Mixed:** 2< R < 5

R= (ALT/ULN)/ (Alk / ULN)

(J Hepatol 1990; 11: 272)

Acute hepatitis: Differential Dx

Ultrasound/ CT

– – –

Viral
(A, B, C, CMV, EBV
HEV, HSV)

Autoimmune
(SPEP, ANA, SmAb)

Ischemia
(History, 2D-Echo)

Metabolic
(Iron, TIBC, ferritin,
ceruloplasmin, SPEP)

Drug

+ +

NAFLD

Mass
(AFP, MRI)

Biliary
(ERCP)

Observe/ biopsy

Idiopathic Hepatitis

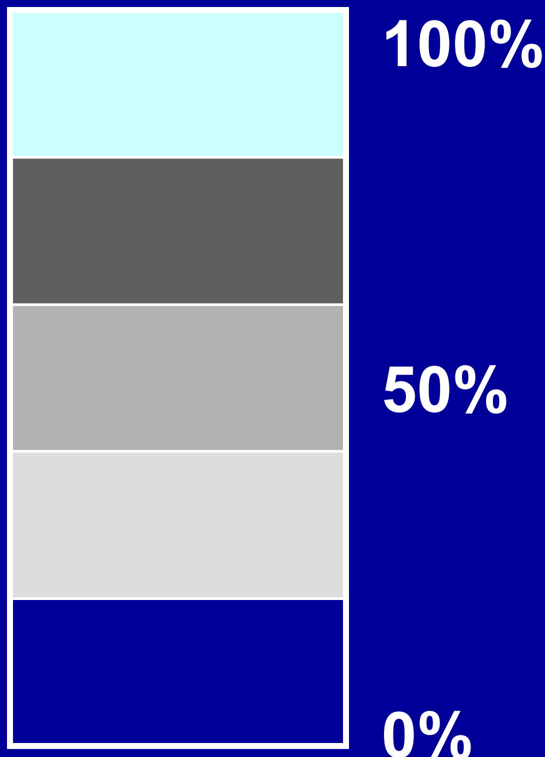
- 22 hospitalizations/ 1,000,000 medicaid-pt yrs for idiopathic acute hepatitis
  - HCV not excluded ('80-'87) *
- Acute "non-A, non-B" hepatitis
  - 5- 8% of post-Txf hepatitis
- 17% of ALF is indeterminate

(Arch Intern Med 1993; 153: 1331)

DILI Diagnosis

- DILI is a diagnosis of exclusion based on circumstantial evidence due to lack of objective confirmatory lab test, rechallenge, or "GOLD" standard
  - Requires a high index of suspicion
- DILI diagnosis is usually retrospective
  - Exclude other causes
  - Dechallenge requires follow-up

ADR: Causality Assessment



- Definite
- Highly probable
- Probable
- Possible
- Unlikely
- Excluded / other

DILI: Causality Assessment

- Generic instruments
  - WHO
  - Bayesian
- Liver specific
  - Expert opinion
  - Roussel Uclaf Causality Assessment Method (RUCAM) ‘89
  - Clinical Diagnostic Scale (CDS) ‘97
  - DILIN approach

DILI Causality Instrument

- Sensitive
- Specific- Low probability in non-drug cases
- Reproducible
- Content validity- weighting is evidence based
- Criterion validity- "Gold Standard" expert panel
- Discrimination- a semi-quantitative estimate
- Validated in independent groups
- Generalizability- Young vs old, mild vs severe, hepatocellular vs cholestatic, normal vs abnormal baseline LFT's
- Ease of use

RUCAM

• Temporal relationship	(0 to 2)
• Course	(-2 to 3)
• Risk factors	(0 to 2)
• Concomitant drug	(0 to -3)
• Non-drug causes	(-3 to 2)
• Prior reports/ information	(0 to 2)
• Re-challenge	(-2 to 3)

Score (-8 to 14)

Highly probable >8 Possible 3-5 Excluded ≤0
Probable 6-8 Unlikely 1-2

J Clin Epidemiol 1993;46:1323-1330

RUCAM limitations

- Ambiguous instructions
  - Criteria for competing cause/drug not clear
- Onset > 30 days after d/c (e.g. Augmentin)
- Derived from expert opinion rather than prospectively collected data set
  - Limited risk factors
  - Overweighting of rechallenge

Clinical Diagnostic Scale

- Temporal association
  - From initiation (1 to 3)
  - From cessation (-3 to 3)
  - Normalization (0 to 3)
- Non-drug causes (-3 to 3)
- Extrahepatic manifestations (0 to 3)
- Rechallenge (0 to 3)
- Prior reports (-3 to 2)

Score (- 9 to 20)

Definite > 17
Probable 14-17
Possible 10-13
Unlikely 6-9
Excluded < 6

(Hepatology 1997;26: 664)

228 Spanish cases ’94-’00
Gold standard: Expert panel

	CDS				
RUCAM	**Exclude**	**Unlike**	**Poss**	**Prob**	**Def**
Exclude	21	2			
Unlike	4	3			
Poss		8	1		
Prob	1	30	43	16	
Definite		5	40	53	1

* 30 non-drug cases

K = 0.28

(Hepatology 2001; 33: 123)

CDS vs RUCAM

- RUCAM performed better overall
- CDS performed poorly if
  - Delayed onset (> 15 days)
  - Prolonged recovery (Cholestasis)
  - ALF/ Death (6%)
    - CDS: 6 possible 7 unlikely/ excluded
    - RUCAM: 6 definite 6 probable 1 possible
  - Idiosyncratic (75%)

(Hepatology 2001; 33: 123)

Drug Induced Liver Injury Network Objectives

- To collect biological samples from bonafide cases and controls to study pathogenesis using biochemical, serological and genetic techniques

- To develop standardized instruments, definitions, and terminology for drug and CAM induced liver injury

DILIN Prospective study

- Inclusion criteria
  - Age > 2
  - Liver injury due to a drug or CAM within 6 months of presentation
  - On 2 consecutive blood draws
    - AST/ ALT > 5 X ULN or baseline
    - Alk phos > 2X ULN or baseline
    - T bilirubin > 2.5 mg/dl

DILIN Prospective study

- Exclusion criteria
  - Acetaminophen hepatotoxicity
  - Prior liver transplant
  - Pre-existing AIH, PSC, PBC which may confound ability to diagnose DILI
    - Chronic HBV, HCV allowed

DILIN: Baseline visit

- **All patients**
- HAV (IgM)
- HBsAg, HBcAb
- HCV ab
- ANA, SmAb
- Monospot
- HIV ab
- Liver ultrasound
- Ceruloplasmin (< 50)
- AMA (Cholestatic)

- **Chronic HBV**
- HBV DNA
- HBeAg, HBeAb
- HDV Ab

- **Chronic HCV**
- HCV RNA

DILIN Causality assessment

- Causality committee
  - 5 site PI's, 1 DCC, 1 NIH
- 3 independent reviewers per case
  - Site PI and 2 others
  - Review clinical narrative, subset CRF, labs
  - Assess causality, RUCAM, Data completeness checklist
- Conference call if not unanimous

DILIN Clinical Narrative

- Presentation
- Detailed medication hx/ compliance
- Concomitant meds/ CAM
- PMH
- Soc/ fam hx
- Physical exam
- Liver enzymes & labs
- Diagnostic tests
  - Central path review
- Outcome

DILIN Causality assessment

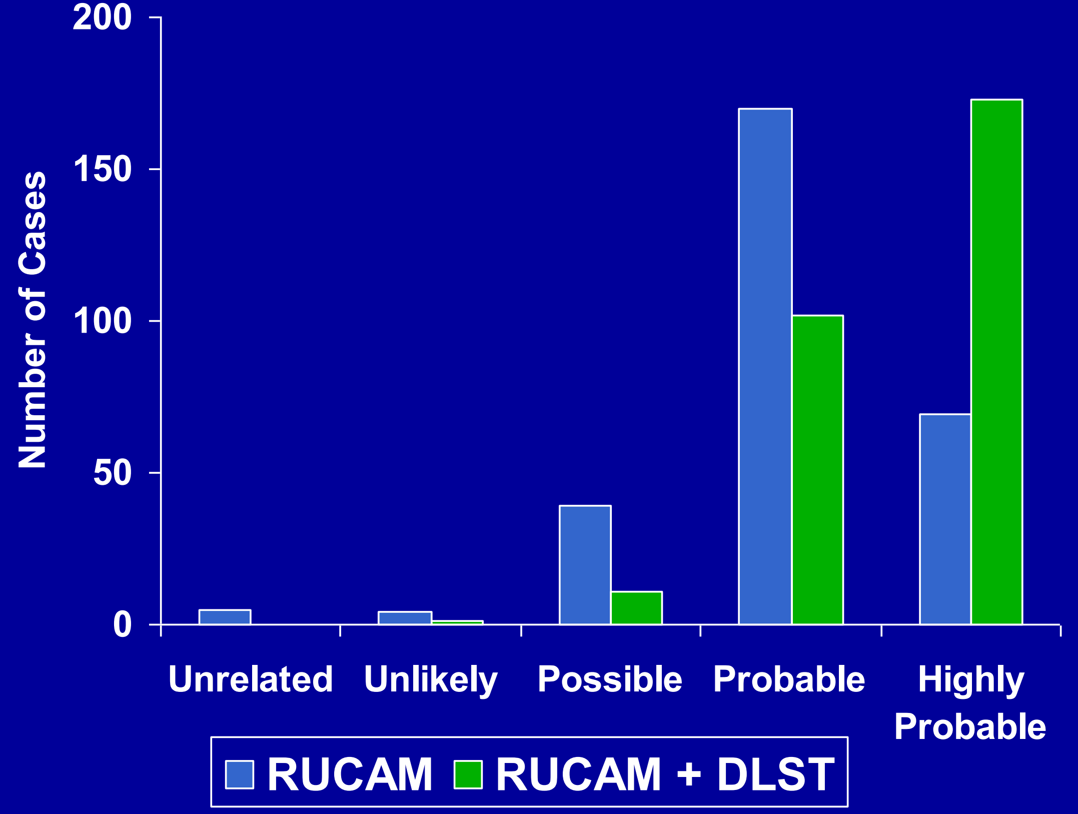
Likelihood	Category
> 95%	Definite
75 -95%	Likely
50 -75%	Probable
25 -50%	Possible
< 25%	Unlikely

Causality Assessment in 2005

- Need Bonafide cases of DILI
  - ? Features ? Risk factors ? Mechanism ? Outcome
  - Surveillance system
- Improved causality assessment instruments
  - Key data, sensitive/ specific, validated
  - User friendly, widely available
- Objective lab test to confirm diagnosis
  - ? Lymphocyte proliferation assays/ biomarker
  - ? Genetic susceptibility test
- Reference database for DILI

287 Japanese DILI cases ’78-’02

55% hepatocellular 24% mixed 22% cholestatic



- Latency > 15 or 30 d changed, other drugs omitted,
- + DLST = +2 Eosinophils > 6% = + 1

(Hep Research 2003: 27: 191)

Fulminant DILI in the US

- 138 DILI ALF LT recipients (15%) ’90-’02
  - Mean age 35 75% female 70% Cau 26% AA
    - 17% INH 9 % PTU 7% dilantin 7% valproate
- 40 DILI (13%) ALFSG ’98-’03
  - Med age 41 72% female 58% Cau
    - 27% antibiotics 10% troglitazone 10% bromfenac
    - 25% spont survival 53% transplant

(Ann Intern Med 2002: 137: 947)
(Liver Transp 2004; 10: 1018)